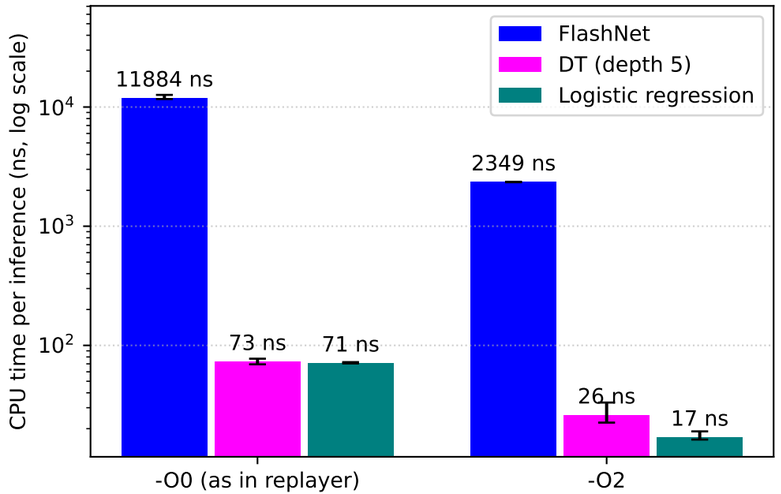
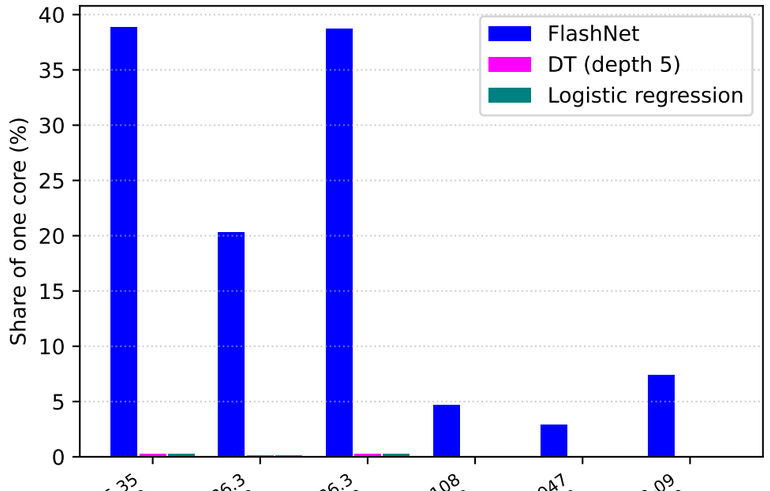
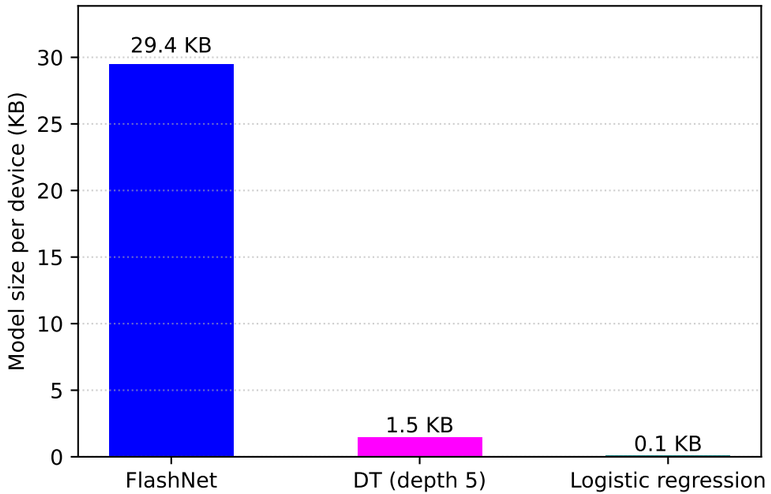

(a) Inference cost (bars: mean, whiskers: min–max
across 6 traces × 2 devices)
CPU time per inference (ns, log scale)
FlashNet
DT (depth 5)
Logistic regression
11884 ns
73 ns
71 ns
2349 ns
26 ns
17 ns
-O0 (as in replayer)
-O2


(b) Estimated CPU share of inference while replaying
(both clients, -O0; = ns/inf × read IOPS)
Share of one core (%)
FlashNet
DT (depth 5)
Logistic regression
ali.iops_p100.alibaba_9086.35
rerate_4.00.resize_4.00
ali.offset_p100.alibaba_9086.3
rerate_2.00.resize_2.00
ali.offset_p100.alibaba_9086.3
rerate_4.00
msr.overall_p100.src1_0.2108
rw_ratio_40_60.rerate_8.00.resize_2.00
msr.write_p100.src1_0.3047
rw_ratio_40_60.rerate_8.00.resize_2.00
tencent.write_p100.2018-10-09
rerate_0.50


(c) Model memory footprint
(weights / tree arrays, from compiled object)
Model size per device (KB)
29.4 KB
1.5 KB
0.1 KB
FlashNet
DT (depth 5)
Logistic regression